Classical theories of information science

信息度量
Information Measures

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6. Maximum Entropy Distributions

Maximum Entropy Distributions

In Theorem 2.43, we have proved that for any random variable X,

$$H(X) \leq \log |\mathcal{X}|, \tag{2.199}$$

with equality when X is distributed uniformly over $\mathcal{X}$. In this section, we revisit this result in the context that X is a real random variable.

To simplify our discussion, all the logarithms are in the base e. Consider the following problem:

Maximize $H(p)$ over all probability distributions p defined on a countable subset $\mathcal{S}$ of the set of real numbers, subject to

$$\sum_{x \in \mathcal{S}_p} p(x) r_i(x) = a_i \quad \text{for } 1 \leq i \leq m, \tag{2.200}$$

where $\mathcal{S}_p \subset \mathcal{S}$ and $r_i(x)$ is defined for all $x \in \mathcal{S}$.

The following theorem renders a solution to this problem.

Maximum Entropy

$$\sum_{x \in \mathcal{S}_p} p(x) r_i(x) = a_i \quad \text{for } 1 \le i \le m, \tag{2.200}$$

Theorem 2.50. *Let*

$$p^*(x) = e^{-\lambda_0 - \sum_{i=1}^{m} \lambda_i r_i(x)} \tag{2.201}$$

for all $x \in \mathcal{S}$, where $\lambda_0, \lambda_1, \cdots, \lambda_m$ are chosen such that the constraints in (2.200) are satisfied. Then p^* maximizes $H(p)$ over all probability distribution p on $\mathcal{S}$, subject to the constraints in (2.200).

Proof of max entropy

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$$\sum_{x \in \mathcal{S}_p} p(x) r_i(x) = a_i \quad \text{for } 1 \le i \le m, \tag{2.200}$$

Proof. For any p satisfying the constraints in (2.200), consider

$$\begin{aligned}
&H(p^*) - H(p) \\
&= -\sum_{x \in \mathcal{S}} p^*(x) \ln p^*(x) + \sum_{x \in \mathcal{S}_p} p(x) \ln p(x) && (2.202) \\
&= -\sum_{x \in \mathcal{S}} p^*(x) \left(-\lambda_0 - \sum_i \lambda_i r_i(x) \right) + \sum_{x \in \mathcal{S}_p} p(x) \ln p(x) && (2.203) \\
&= \lambda_0 \left(\sum_{x \in \mathcal{S}} p^*(x) \right) + \sum_i \lambda_i \left(\sum_{x \in \mathcal{S}} p^*(x) r_i(x) \right) + \sum_{x \in \mathcal{S}_p} p(x) \ln p(x) && (2.204) \\
&= \lambda_0 \cdot 1 + \sum_i \lambda_i a_i + \sum_{x \in \mathcal{S}_p} p(x) \ln p(x) && (2.205) \\
&= \lambda_0 \left(\sum_{x \in \mathcal{S}_p} p(x) \right) + \sum_i \lambda_i \left(\sum_{x \in \mathcal{S}_p} p(x) r_i(x) \right) + \sum_{x \in \mathcal{S}_p} p(x) \ln p(x) && (2.206)
\end{aligned}$$

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for all $x \in \mathcal{S}$, where $\lambda_0, \lambda_1, \cdots, \lambda_m$ are chosen such that the constraints in (2.200) are satisfied. Then p^* maximizes $H(p)$ over all probability distribution p on $\mathcal{S}$, subject to the constraints in (2.200).

$$\sum_{x \in \mathcal{S}_p} p(x) r_i(x) = a_i \quad \text{for } 1 \le i \le m, \tag{2.200}$$

$$= -\sum_{x \in \mathcal{S}_p} p(x) \left(-\lambda_0 - \sum_i \lambda_i r_i(x) \right) + \sum_{x \in \mathcal{S}_p} p(x) \ln p(x) \tag{2.207}$$

$$= -\sum_{x \in \mathcal{S}_p} p(x) \ln p^*(x) + \sum_{x \in \mathcal{S}_p} p(x) \ln p(x) \tag{2.208}$$

$$= \sum_{x \in \mathcal{S}_p} p(x) \ln \frac{p(x)}{p^*(x)} \tag{2.209}$$

$$= D(p \| p^*) \tag{2.210}$$

$$\ge 0. \tag{2.211}$$

In the above, (2.207) is obtained from (2.203) by replacing $p^*(x)$ by $p(x)$ and $x \in \mathcal{S}$ by $x \in \mathcal{S}_p$ in the first summation, while the intermediate steps (2.204) to (2.206) are justified by noting that both p^* and p satisfy the constraints in (2.200). The last step is an application of the divergence inequality (Theorem 2.31). The proof is accomplished. □

Proof of max entropy

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Theorem 2.50. *Let*

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Proof. For any p satisfying the constraints in (2.200), consider

$$\begin{aligned}
&H(p^*) - H(p) \\
&= -\sum_{x\in\mathcal{S}} p^*(x) \ln p^*(x) + \sum_{x\in\mathcal{S}_p} p(x) \ln p(x) && (2.202)\\
&= -\sum_{x\in\mathcal{S}} p^*(x) \left(-\lambda_0 - \sum_i \lambda_i r_i(x)\right) + \sum_{x\in\mathcal{S}_p} p(x) \ln p(x) && (2.203)\\
&= \lambda_0 \left(\sum_{x\in\mathcal{S}} p^*(x)\right) + \sum_i \lambda_i \left(\sum_{x\in\mathcal{S}} p^*(x) r_i(x)\right) + \sum_{x\in\mathcal{S}_p} p(x) \ln p(x) && (2.204)\\
&= \lambda_0 \cdot 1 + \sum_i \lambda_i a_i + \sum_{x\in\mathcal{S}_p} p(x) \ln p(x) && (2.205)\\
&= \lambda_0 \left(\sum_{x\in\mathcal{S}_p} p(x)\right) + \sum_i \lambda_i \left(\sum_{x\in\mathcal{S}_p} p(x) r_i(x)\right) + \sum_{x\in\mathcal{S}_p} p(x) \ln p(x) && (2.206)\\
&= -\sum_{x\in\mathcal{S}_p} p(x) \left(-\lambda_0 - \sum_i \lambda_i r_i(x)\right) + \sum_{x\in\mathcal{S}_p} p(x) \ln p(x) && (2.207)\\
&= -\sum_{x\in\mathcal{S}_p} p(x) \ln p^*(x) + \sum_{x\in\mathcal{S}_p} p(x) \ln p(x) && (2.208)\\
&= \sum_{x\in\mathcal{S}_p} p(x) \ln \frac{p(x)}{p^*(x)} && (2.209)\\
&= D(p\|p^*) && (2.210)\\
&\ge 0. && (2.211)
\end{aligned}$$

In the above, (2.207) is obtained from (2.203) by replacing $p^*(x)$ by $p(x)$ and $x \in \mathcal{S}$ by $x \in \mathcal{S}_p$ in the first summation, while the intermediate steps (2.204) to (2.206) are justified by noting that both p^* and p satisfy the constraints in (2.200). The last step is an application of the divergence inequality (Theorem 2.31). The proof is accomplished. □

Corollary 2.51. *Let p^* be a probability distribution defined on $\mathcal{S}$ with*

$$p^*(x) = e^{-\lambda_0 - \sum_{i=1}^{m} \lambda_i r_i(x)} \tag{2.212}$$

for all $x \in \mathcal{S}$. Then p^* maximizes $H(p)$ over all probability distribution p defined on $\mathcal{S}$, subject to the constraints

$$\sum_{x \in \mathcal{S}_p} p(x) r_i(x) = \sum_{x \in \mathcal{S}} p^*(x) r_i(x) \quad \text{for } 1 \le i \le m. \tag{2.213}$$

Examples

Example 2.52. Let $\mathcal{S}$ be finite and let the set of constraints in (2.200) be empty. Then

$$p^*(x) = e^{-\lambda_0}, \tag{2.214}$$

a constant that does not depend on x. Therefore, p^* is simply the uniform distribution over $\mathcal{S}$, i.e., $p^*(x) = |\mathcal{S}|^{-1}$ for all $x \in \mathcal{S}$. This is consistent with Theorem 2.43.

Examples

Theorem 2.50. *Let*

$$p^*(x) = e^{-\lambda_0 - \sum_{i=1}^m \lambda_i r_i(x)} \tag{2.201}$$

for all $x \in \mathcal{S}$, where $\lambda_0, \lambda_1, \cdots, \lambda_m$ are chosen such that the constraints in (2.200) are satisfied. Then p^* maximizes $H(p)$ over all probability distribution p on $\mathcal{S}$, subject to the constraints in (2.200).

Example 2.53. Let $\mathcal{S} = \{0, 1, 2, \cdots\}$, and let the set of constraints in (2.200) be

$$\sum_x p(x)x = a, \tag{2.215}$$

where $a \geq 0$, i.e., the mean of the distribution p is fixed at some nonnegative value a. We now determine p^* using the prescription in Theorem 2.50. Let

$$q_i = e^{-\lambda_i} \tag{2.216}$$

for $i = 0, 1$. Then by (2.201),

$$p^*(x) = q_0 q_1^x. \tag{2.217}$$

Evidently, p^* is a geometric distribution, so that

$$q_0 = 1 - q_1. \tag{2.218}$$

Finally, we invoke the constraint (2.200) on p to obtain $q_1 = (a+1)^{-1}$. The details are omitted.

Examples

Example 12.2.1 (*One-dimensional gas with a temperature constraint*) Let the constraints be $EX = 0$ and $EX^2 = \sigma^2$. Then the form of the maximizing distribution is

$$f(x) = e^{\lambda_0 + \lambda_1 x + \lambda_2 x^2}. \tag{12.13}$$

To find the appropriate constants, we first recognize that this distribution has the same form as a normal distribution. Hence, the density that satisfies the constraints and also maximizes the entropy is the $\mathcal{N}(0, \sigma^2)$ distribution:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}. \tag{12.14}$$

Thank you!